

TIC 86880482

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.828371 d
Epoch (t0) 1654.8347 BTJD
Duration 0.1700 d
Depth 3.588 % (35,885 ppm)
SNR 150.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

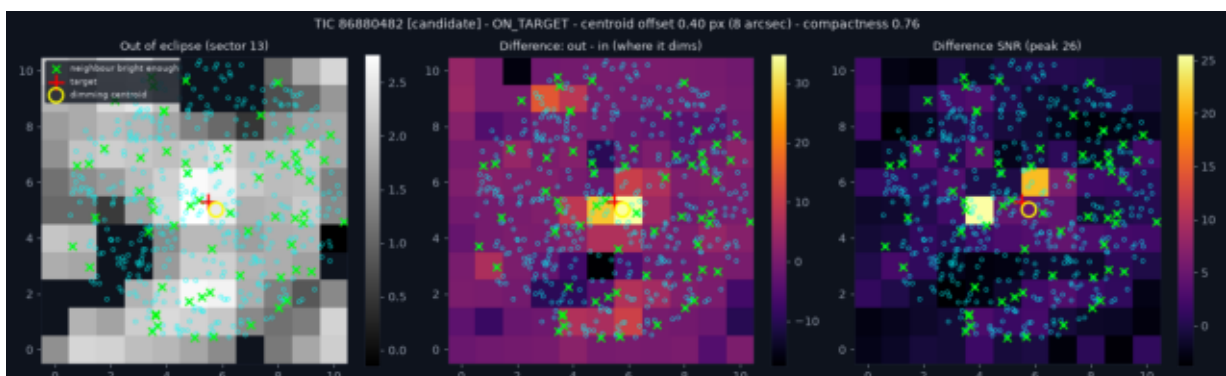
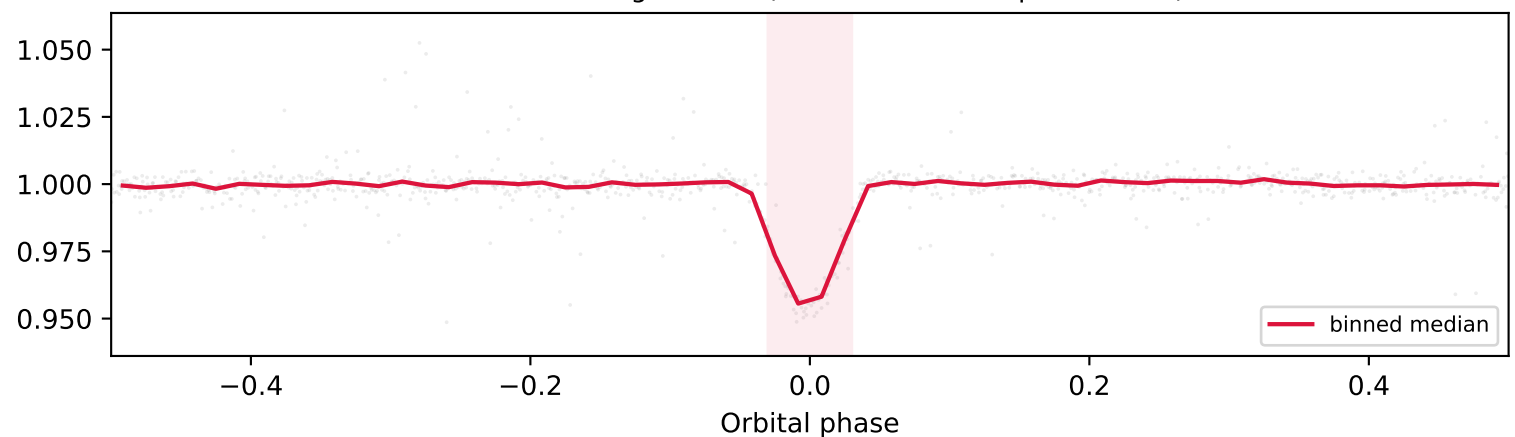
Eclipses observed 8 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 89.8 (gate ≥ 7.0)
Odd/even z-score 0.23 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 87.1 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 25.59 (gate ≥ 5.0) Compactness 0.76 (gate ≥ 0.50)
Dimming centroid offset 0.40 px (8 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 86880494
0.51 px away, dTmag +2.45 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.